O'REILLY

# 19 Intelligent Automation Systems (IAS) and Its Application in Neuroscience

**Bikram Kar[1*] and Amit Kumar[2]**

*[1]Asansol Engineering College, Asansol, Paschim Barddhaman, West Bengal, India*

*[2]Sanaka Educational Trust's Group of Institutions, Durgapur, West Bengal, India*

***Abstract***

Intelligent automation systems (IAS) are smart machines that are transforming the way we study neuroscience. These systems use artificial intelligence and robotics to conduct complex neuroscience tests and evaluate data with minimal human intervention. IAS can rapidly assess ner-

vous system connections, analyze impulses in nerve cells, and view comprehensive images of neural tissues. By doing activities that humans find difficult or time-consuming, IAS allows scientists to investigate the neural system more accurately and on a bigger scale. These devices can interpret massive amounts of neuroscience data, allowing researchers to learn new things about how nerves and the brain operate together. Using IAS has resulted in significant improvements in comprehending the neurological system and related illnesses. As these systems improve, they may contribute to the discovery of new treatments for neurological diseases. However, utilizing IAS enables scientists to consider how to interpret data and how this improves their position in neuroscience research.

***Keywords*:** Neuroscience, intelligent automation systems (IAS), machine learning, neural interfaces, human-AI interaction, neurotechnology

# 19.1 Introduction

## 19.1.1 Defining Neuroscience and Intelligent Automation Systems

Neuroscience is the multidisciplinary scientific study of the nervous system, with a primary focus on the brain. It encompasses diverse research areas that investigate various aspects of neural structure, function, development, and pathology. The main branches include neuroanatomy, which studies the structure and organization of the nervous system from individual neurons to larger neural pathways and cortical regions; neurophysiology, which examines the functional properties and mechanisms governing neural activity, such as action potentials and neurotransmitter dynamics; molecular and cellular neuroscience, investigating the molecular and cellular components underlying neural function like receptors and signaling pathways; systems neuroscience, studying how neural circuits integrate information to produce behaviors and cognitive functions; cognitive neuroscience, exploring the neural mechanisms of higher-order processes like perception and memory through combined neuroscience and psychological approaches; and computational neuroscience, developing models and simulations to understand neural computation principles. Neuroscientists utilize techniques like neuroimaging, electrophysiology,

optogenetics, and computational modeling to unravel the complexities of the nervous system.

Intelligent automation systems (IAS) refer to integrating advanced technologies like artificial intelligence (AI), machine learning (ML), robotics, and automation to optimize and automate tasks. IAS mimic and augment human cognitive abilities, enabling complex decision-making, problem-solving, and execution automation. Key IAS components include AI algorithms exhibiting human-like intelligence for perception, learning, and reasoning; ML models that learn and improve from data; robotics involving designing and operating agents that perceive environments and act [1]; and automation technologies controlling processes with minimal human intervention for efficiency [2]. IAS leverage these components to automate complex tasks, make data-driven decisions, and adapt, enhancing productivity and decision-making across domains like industrial automation, healthcare, finance, and transportation.

### 19.1.2 Importance of Bridging Neuroscience and Intelligent Automation

The convergence of neuroscience and intelligent automation holds immense potential, promising to revolutionize numerous fields and unlock new frontiers in technology and scientific understanding. By bridging these two domains, researchers and engineers can leverage the insights and principles derived from the study of the brain to develop more advanced, efficient, and human-centric automation systems.

**Bioinspired Algorithms and Architectures:**

Neuroscience research has revealed many remarkable information-processing capabilities of the brain that can inspire the development of novel algorithms and architectures for IAS. Some key examples include:

- **Artificial Neural Networks:** The artificial neural network models used widely in deep learning are directly inspired by the networked architecture of the brain's neurons and their connections (synapses). However, modern neural nets are still simplistic compared to biological neural networks. Insights from neuroscience on more complex

neural dynamics, plasticity rules, neuromodulation, and principles like sparse coding could lead to more powerful artificial neural architectures.

- **Reservoir Computing:** This biologically inspired computational paradigm is based on neuroscientific principles of exploiting transient dynamics in recurrent neural networks with fixed, random connectivity. It has been applied to tasks like speech recognition by leveraging the brain's ability to integrate inputs over time into rich dynamical patterns. Insights from neuroscience on neural coding could refine and extend reservoir computing approaches.
- **Spiking Neural Networks:** Unlike traditional neural nets, spiking neural networks more faithfully model how real neurons communicate via discrete spike timing events. Taking inspiration from neuroscience on neural coding strategies like firing rate codes and temporal codes, spiking networks aim to achieve similar computational power as the brain using sparse, event-based representations.
- **Neuromorphic Computing:** This involves designing electronic circuits and hardware architectures that mimic neuro-biological structures and information-processing principles. For example, IBM's TrueNorth and Intel's Loihi chips implement multi-compartment neuron models and synaptic plasticity rules inspired by neuroscience, enabling low-power, highly parallel, fault-tolerant computing.
- **Developmental/Evolutionary Algorithms:** By capturing insights from neuroscience on processes like neurogenesis, neural pruning, neuromodulation, and evolution of neural circuits, new developmental and evolutionary algorithms could be created that exhibit increasing complexity and continuous learning and adaptation properties akin to biological brains.
- **Self-Organizing Maps/Neural Fields:** These algorithmic models draw inspiration from neuroscientific concepts of topographic mapping and lateral inhibition/competition principles underlying brain functions like visual processing and sensorimotor integration.

Overall, by rigorously studying organizational principles, neural coding strategies, plasticity mechanisms, neurotransmitter systems, and developmental trajectories in the brain, novel and more capable computational

algorithms and architectural paradigms can be derived for intelligent systems design.

**Enhancing Human-Machine Interaction:**

Understanding the neural mechanisms of perception, cognition, and communication can inform the design of intelligent interfaces and interactions between humans and machines. This knowledge can contribute to the development of more intuitive, personalized, and userfriendly automation systems that can seamlessly integrate with human operators and users.

Here are some details on how understanding neural mechanisms from neuroscience can enhance human-machine interaction, with examples:

- **Multimodal Interfaces:** Insights from neuroscience on how the brain integrates inputs from multiple sensory modalities like vision, audition, and touch can guide the design of intelligent multimodal interfaces. For example, studying neural activation patterns in multimodal association cortices could reveal principles for optimally combining video, audio, haptic, and other channels to create intuitive and natural user experiences.
- **Adaptive User Interfaces:** Knowledge of the neural underpinnings of cognitive processes like attention, working memory, cognitive load, and fatigue could enable adaptive interfaces that automatically adjust information complexity and presentation based on the user's current mental state decoded from brain activity patterns. This could prevent cognitive overload and frustration.
- **Neural Language Interfaces:** By mapping the neural coding of language in areas like Wernicke's and Broca's areas, brain-computer interfaces (BCIs) could eventually translate neural signals into natural language commands to control machines, or render machine outputs into comprehensible neural responses. Companies like Kernel are already exploring such language-brain interfaces.
- **Neurofeedback Interaction:** Intelligent systems could provide users with realtime neurofeedback based on decoding their neural states related to factors like engagement, flow, and frustration. User neural sig-

natures of desired cognitive states could be used to dynamically adapt and optimize system interactions.

- **Affective Computing:** Understanding the neural circuits involved in emotional processing, motivation, and social cognition from neuroscience could lead to affective computing systems that can detect, interpret, and respond appropriately to human emotional/mental states during interactions.
- **Neural Plasticity-Based Customization:** As machines increasingly become longterm companions, understanding neural plasticity principles could allow them to customize their interactions in a way that gets imprinted or encoded in the user's brain for intuitive ease-of-use through neuroadaptive updates.

**Assistive Technologies:**

Neuroscience-informed IAS can play a pivotal role in developing advanced assistive technologies for individuals with disabilities or neurological disorders. By leveraging insights into the brain's plasticity and compensatory mechanisms, these systems can be designed to help restore, enhance, or compensate for impaired cognitive, sensory, or motor functions.

Here is a detailed look at how neuroscience-informed IAS can advance assistive technologies for individuals with disabilities or neurological disorders:

- **Brain-Computer Interfaces (BCIs):** By decoding neural activity patterns, BCIs can enable communication and control for those with severe motor impairments like locked-in syndrome. Neuroscience insights into the neural coding of intentions, movements, and sensory processing are allowing more naturalistic and high-bandwidth BCIs. For example, reading out neural population codes for intended arm movements could seamlessly control a robotic prosthetic limb.
- **Cognitive Training and Enhancement:** Combining neurofeedback techniques that provide real-time feedback on specific neural activation patterns, with AI systems that personalize cognitive training regimens, could enhance abilities like attention, working memory, or decision-making in those with ADHD, autism, or neurological deficits.

Neural plasticity principles guide developing optimal neurofeedback and cognitive restructuring protocols.

- **Sensory Substitution and Augmentation:** For those with visual or auditory impairments, wearable devices that encode visual/auditory information into alternate senses like touch or electro tactile feedback on the tongue could provide sensory augmentation and substitution. Studying cross-modal neural reorganization and plasticity after sensory loss guides designing intuitive sensory neuroprosthetics.
- **Intelligent Neuromodulation Therapies:** AI and neuroscience are converging to enable intelligent neuromodulation therapies like personalized deep brain stimulation for Parkinson's or responsive neurostimulation to prevent epileptic seizures. ML on neural biomarkers and modeling neural dynamics allow for optimizing stimulation protocols for precise therapeutic neuromodulation.
- **Neural Rehabilitation and Restoration:** After stroke, trauma, or neurodegenerative diseases, neural plasticity and reorganization hold the key to rehabilitation. AI algorithms can analyze vast multimodal data (neuroimaging, electrophysiology, behavioral) to derive patient-specific neural signatures of recovery and create optimized therapeutic regimens involving activities, neurofeedback, non-invasive brain stimulation, and even neural implants.

Overall, by reverse-engineering neural coding strategies, plasticity rules, and reorganization principles from neuroscience and combining this knowledge with intelligent technologies like BCIs, adaptive AI systems, robotics, and neuromodulation devices, a new era of powerful and personalized assistive neurotechnology is emerging to help restore and augment human cognition and abilities.

**Neurorehabilitation and Neurotherapeutics:**

The fusion of neuroscience and intelligent automation can contribute to the advancement of neurorehabilitation and neurotherapeutic interventions. IAS can be employed to design personalized rehabilitation programs, monitor neural activity, and provide real-time feedback to patients and clinicians, ultimately improving the efficacy of treatments for various neurological conditions.

Here is a detailed look at how the fusion of neuroscience and IAS can advance neurorehabilitation and neurotherapeutic interventions:

- **Personalized Rehabilitation Programs:** Analyzing multi-modal data like neuroimaging, electrophysiology, genetics, and behavioral assessments using ML algorithms can reveal unique neural signatures and biomarkers associated with an individual's brain injury or disorder. These personalized neural profiles can then guide the design of customized rehabilitation programs tailored to target the specific neural circuits and mechanisms impacted in that patient. For example, after a stroke, the program may focus on retraining motor functions by stimulating the brain's residual motor pathways through neurofeedback and activity regimens optimized for that patient's neural reorganization patterns.
- **Neural Activity Monitoring:** Wearable brain monitoring devices like high-density EEG or functional near-infrared spectroscopy (fNIRS) headsets can continuously track a patient's neural activation patterns during rehabilitation exercises or daily activities. AI systems can then decode this neural data stream in real time to assess the patient's cognitive state, level of effort, and degree of neural plasticity being engaged. This neural readout can adapt and adjust the rehabilitation tasks accordingly for maximum benefit.
- **Real-Time Neurofeedback:** By instantly decoding a patient's neural patterns, IAS can provide them with real-time neurofeedback and guidance during rehabilitation tasks. Virtual reality environments can be rendered to optimally stimulate the desired neural circuitry based on the decoded neural signals. Neurofeedback has been shown to accelerate functional gains by consciously driving neuroplasticity. Additionally, similar neural feedback loops can directly control therapeutic neurostimulation parameters delivered to the patient's brain in a closed-loop system.
- **Simulated Neural Dynamics:** Computational models simulating the biophysical dynamics of neurons, neural networks, and brain regions can create *in silico* "brain emulations." By integrating patient neural data into these simulations, clinicians can run virtual trials to predict optimal stimulation sites/parameters and test potential rehabilitation strategies in a simulated environment before clinical implementation.

This shortens trial-and-error and maximizes positive neurological outcomes.

Overall, by combining neural decoding algorithms, brain simulation models, intelligent control systems, and advanced neural interfaces, the neuroscience-automation nexus is opening up new possibilities for precise, personalized, and optimized delivery of neurotherapeutic and neurorehab interventions tuned to drive each patient's unique neural recovery landscape.

**Ethical Considerations:**

As the field of intelligent automation continues to evolve, it is essential to consider the ethical implications of these technologies, particularly in relation to their impact on human well-being, privacy, and societal dynamics. Neuroscience can provide valuable insights into the neural basis of moral decision-making, empathy, and consciousness, informing the development of ethical frameworks and guidelines for the responsible deployment of IAS.

Here are some key ethical considerations surrounding the convergence of neuroscience and IAS:

- **Neural Privacy and Data Rights:** As neural interfaces like BCIs and neuroimaging become integrated with IAS, there are serious privacy implications around recording, storing, and utilizing an individual's neural data, which represents their cognitions, emotions, and essence of being. Just like genetic data, strong regulations around neural data privacy, ownership, and consent need to be established. For example, preventing misuse of neural marketing to subliminally influence consumers based on their decoded preferences.
- **Algorithmic Bias and Fair AI:** ML algorithms can inadvertently pick up societal biases encoded in training data. Combining this with neuroscience insights around implicit biases hardwired in human brain networks could lead to discriminatory or unfair AI systems that propagate societal prejudices related to race, gender, age, etc. Techniques from the emerging field of Neural Ethics Trading studying neural representations of moral value could help develop de-biasing methods.

- **Moral Decision-Making:** Understanding the neural basis of moral reasoning from neuroscience could ensure AI systems deployed in law, finance, and healthcare are aligned with human ethical norms. Brain-inspired algorithms trained on neural data representing ethical judgments could aid moral decision-making in autonomous systems. However, simulating the neurological basis of morality *in silico* raises philosophical questions around artificial consciousness.
- **Human-AI Symbiosis:** Continuous interaction with intelligent systems could have unintended effects on neural development and plasticity, especially in children. The degree of human-AI symbiosis needs careful analysis from a neural and cognitive sciences perspective. For example, offloading cognitive tasks to AI could degrade the biological neural architecture involved in those functions.
- **Neurorights and Identity:** Advances in neural engineering and enhancement raise questions around neurorights - the rights around one's neural identity and cognition. For instance, the ethics of corporations or governments enhancing or modifying neural circuits of employees/citizens using neural implants or stimulation raises issues around psychological autonomy.

By drawing insights from cognitive neuroscience on phenomena like moral reasoning, consciousness, cognitive biases, and neural plasticity, ethical frameworks can be established to ensure responsible and equitable development of neuroscience-inspired intelligent automation that respects human privacy, identity, and cognitive liberties.

### 19.1.3 Overview of the Chapter Structure

This chapter aims to explore the synergies between neuroscience and intelligent automation, highlighting the potential benefits, challenges, and future directions of this interdisciplinary field. The chapter is structured as follows:

1. **Understanding Neurosciences:** This section delves into the fundamentals of neuroscience, including the structure and function of neurons, neural networks, and the brain, as well as the key principles of cognitive neuroscience and neural computation. It provides a compre-

hensive overview of the current state of knowledge in the field of neuroscience, laying the groundwork for understanding its potential applications in IAS.

2. **How Neuroscience Can Help in Understanding Intelligent Automation Systems (IAS):** This section examines how neuroscientific insights into cognitive processes, such as perception, attention, memory, and decision-making, can inform the design and development of IAS. It explores the potential of brain-inspired algorithms and models for IAS, discussing how these biologically-based approaches can enhance the capabilities and performance of automation systems.
3. **Application Areas of Neuroscience in IAS:** This section highlights the diverse applications of neuroscience in various domains, including healthcare and medicine, finance and economics, and manufacturing and industry. It showcases how the integration of neuroscience and IAS can lead to groundbreaking advancements in these fields, from personalized medical interventions to intelligent decision-making systems.
4. **Connecting IAS and Neuroscience:** This section delves into the integration of neuroscience and intelligent automation, focusing on topics such as neurofeedback, BCIs, cognitive modeling, human-machine interaction, and neuroergonomics. It explores how these synergies can enhance the seamless collaboration between humans and automation systems, improving overall efficiency, productivity, and user experience.
5. **Challenges and Future Directions:** The final section discusses the ethical and social implications of the intersection between neuroscience and IAS, addressing concerns related to privacy, data protection, algorithmic bias, and the societal impact of these technologies. It also examines the technical challenges that need to be addressed and explores the emerging trends and future directions in this rapidly evolving interdisciplinary field.

Throughout the chapter, relevant case studies, research findings, and technological advancements will be presented to illustrate the practical applications and the transformative potential of the integration between neuroscience and intelligent automation.

# 19.2 Understanding Neurosciences

## 19.2.1 Fundamentals of Neuroscience

### Neurons and Neural Networks

The fundamental building blocks of the nervous system are neurons, which are specialized cells responsible for the transmission and processing of information. Neurons possess a cell body (soma), dendrites for receiving inputs, and an axon for transmitting outputs. These individual neurons form intricate networks, known as neural networks, by connecting and communicating with each other through synapses.

Synapses are the junctions where the axon of one neuron transmits signals to the dendrites or cell body of another neuron. This transmission of signals occurs through the release of chemical neurotransmitters, which bind to receptors on the receiving neuron, triggering an electrical impulse (action potential) that propagates along the neuron's axon.

The complex interconnectivity of neurons within neural networks enables the brain to perform a wide range of cognitive functions, such as perception, memory, learning, and decision-making. The study of the structure, organization, and dynamic properties of these neural networks is central to understanding the fundamental principles of information processing in the brain. Figure 19.1 shows the schematic representation of a neuron and it's key components.

### Brain Structure and Function

The human brain is a remarkably complex and sophisticated organ, comprising various anatomical regions and structures that work in concert to control and coordinate the body's functions. The brain can be broadly divided into the following major regions:

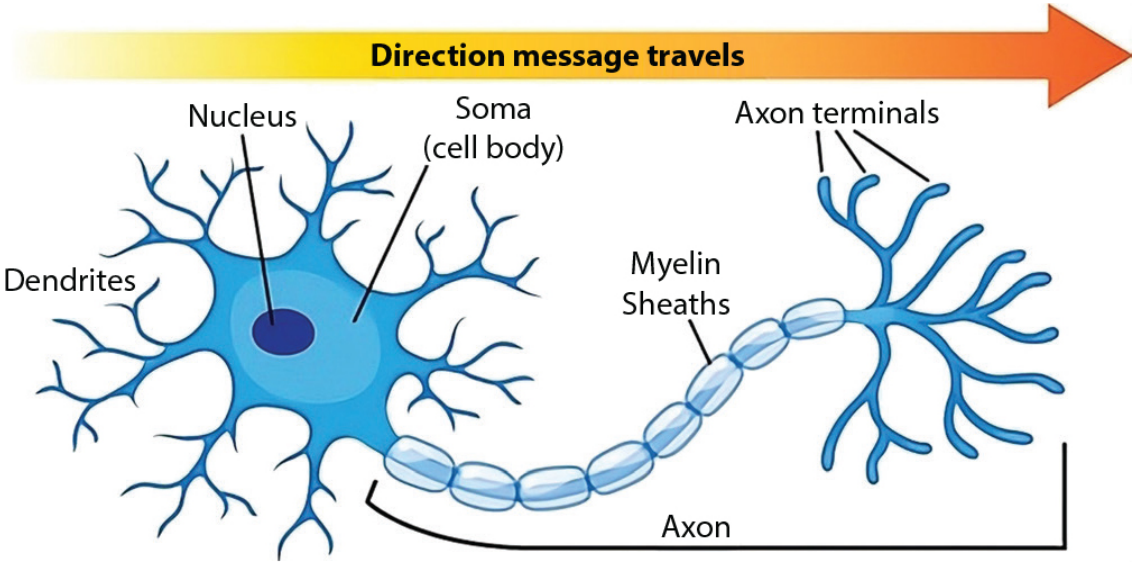



**Figure 19.1** Schematic representation of a neuron and its key components [3].

1. **Cerebrum:** The largest part of the brain, responsible for higher cognitive functions, including conscious thought, language, and voluntary movement.
2. **Cerebellum:** Situated at the back of the brain, the cerebellum plays a crucial role in coordinating and fine-tuning movements, as well as in maintaining balance and posture.
3. **Brainstem:** Connecting the cerebrum and cerebellum to the spinal cord, the brainstem is responsible for regulating vital functions, such as respiration, heart rate, and sleep-wake cycles.
4. **Limbic System:** This collection of interconnected structures, including the hippocampus, amygdala, and hypothalamus, is primarily involved in processing emotions, memories, and motivational drives.

Each of these brain regions is further subdivided into specialized areas, each with distinct functional roles. For instance, the cerebral cortex, the outermost layer of the cerebrum, can be divided into four lobes (frontal, parietal, temporal, and occipital), each associated with specific cognitive and sensory-motor functions.

Understanding the intricate anatomical structure and functional organization of the brain is crucial for neuroscientists to uncover the neural mechanisms underlying complex cognitive processes and behaviors.

### 19.2.2 Cognitive Neuroscience

Cognitive neuroscience is a multidisciplinary field that investigates the neural basis of mental processes, such as perception, attention, memory, language, decision-making, and problem-solving. By integrating knowledge from various disciplines, including psychology, neurobiology, and computer science, cognitive neuroscientists aim to elucidate the relationship between brain structure, function, and cognition. Figure 19.2 represents the major anatomical regions of the human brain.

#### Perception and Attention

Perception is the process by which the brain interprets and organizes sensory information, such as visual, auditory, and somatosensory inputs, to create a meaningful representation of the external world. Researchers in cognitive neuroscience have identified various brain regions and neural mechanisms involved in different aspects of perception, such as the primary sensory cortices, the ventral and dorsal visual pathways, and the attentional networks.

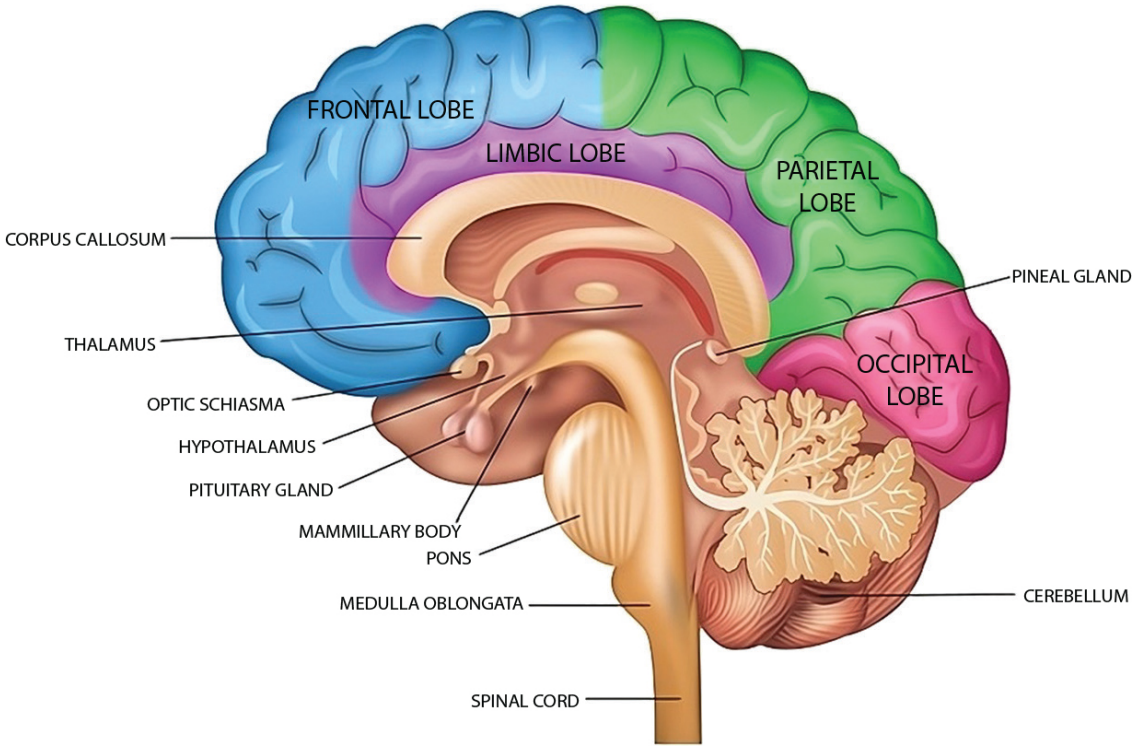



**Figure 19.2** Major anatomical regions of the human brain.

Attention is the cognitive process of selectively focusing on specific information or stimuli while ignoring others. The neural mechanisms of attention involve a complex interplay between the prefrontal cortex, parietal cortex, and subcortical structures, such as the thalamus and basal ganglia. Understanding the neural basis of attention is crucial for designing IAS that can effectively prioritize and process relevant information.

### Memory and Decision-Making

Memory is the ability to encode, store, and retrieve information, and it is a fundamental cognitive function that underpins many higher-order processes. Cognitive neuroscientists have identified the key brain regions involved in different types of memory, such as the hippocampus for declarative memory and the prefrontal cortex for working memory. Insights into the neural mechanisms of memory can inform the development of intelligent systems with enhanced learning and knowledge-retention capabilities.

Decision-making is the cognitive process of selecting the most appropriate course of action from available alternatives. Research in cognitive neuro-

science has revealed the involvement of various brain regions, including the prefrontal cortex, basal ganglia, and limbic system, in different aspects of decision-making, such as value-based, risk-based, and emotion-based decisions. Understanding the neural basis of decision-making can aid in the design of IAS with more robust and adaptive decision-making capabilities.

By integrating the insights gained from these various domains of cognitive neuroscience, researchers can uncover the fundamental principles underlying human cognition and intelligence, which can then be leveraged to enhance the capabilities of IAS.

### 19.2.3 Neural Computation

The field of neural computation focuses on understanding the information-processing capabilities of biological neural networks, such as those found in the brain. At the core of neural computation are the individual neurons and their synaptic connections, which work together to transmit, process, and store information. Computational neuroscience models attempt to capture the complex dynamics of neural activity, such as the generation of action potentials, the modulation of synaptic strengths, and the emergence of population-level neural activity patterns [4]. By developing mathematical and computational models of neural information processing, researchers in this field aim to elucidate the underlying principles that govern how the brain computes. These models have applications in areas like ML, AI, and the development of BCIs [5].

#### Learning and Plasticity

The capacity of the brain to adapt and change in response to experience is a fundamental property known as neuroplasticity. Neural learning involves the modification of the strength and structure of synaptic connections, a process that is believed to underlie the formation of memories and the acquisition of new skills [7]. Mechanisms of synaptic plasticity, such as longterm potentiation and long-term depression, have been extensively studied and are thought to be crucial for the brain's ability to learn and adapt. Beyond synaptic changes, neuroplasticity also encom-

passes other structural and functional alterations in neural circuits, including the formation of new neurons (neurogenesis) and the reorganization of cortical maps [8]. Understanding the neural mechanisms of learning and plasticity is not only important for our fundamental understanding of brain function but also has implications for the development of therapies for neurological and psychiatric disorders. Figure 19.3 represents a simplified diagram of a neural network model, with interconnected nodes representing neurons.

### Neural Encoding and Decoding

The process by which the brain represents and processes sensory information is a central topic in computational neuroscience. Neural encoding refers to the transformation of sensory stimuli into patterns of neural activity, which can be measured as changes in the firing rates or temporal dynamics of individual neurons or populations of neurons. The process of neural encoding and decoding is represented by Figure 19.4. Conversely, neural decoding involves the extraction of information from these neural signals to generate a representation of the external world or to guide behavior. Researchers have made significant progress in developing computational models and experimental techniques to study neural encoding and decoding, with applications in the fields of sensory neuroscience, BCIs, and neuroprosthetics [9]. By better understanding how the brain encodes and decodes information, we can gain insights into the fundamental principles of neural information processing and develop novel technologies to interface with the nervous system.

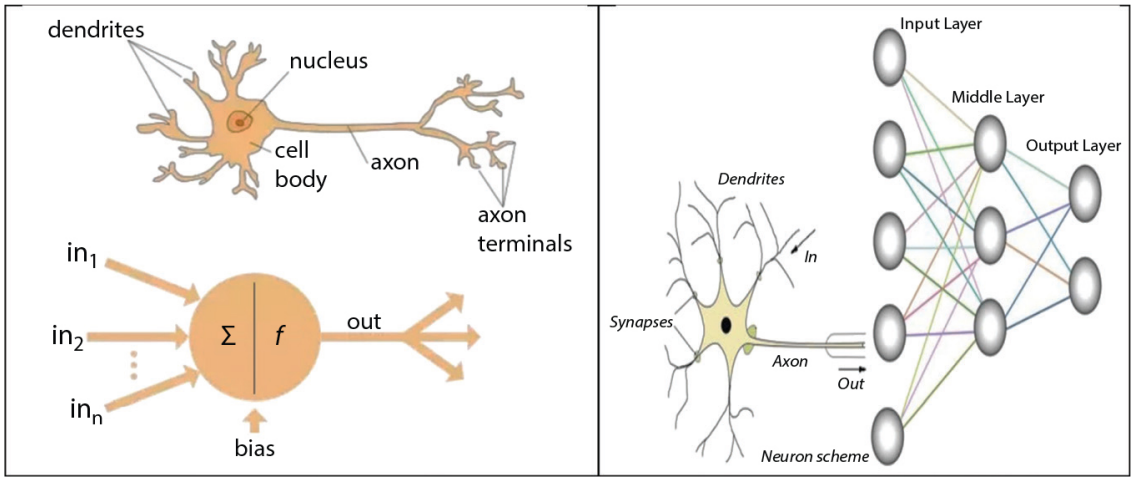



**Figure 19.3** A simplified diagram of a neural network model, with interconnected nodes representing neurons and their synaptic connections [6].

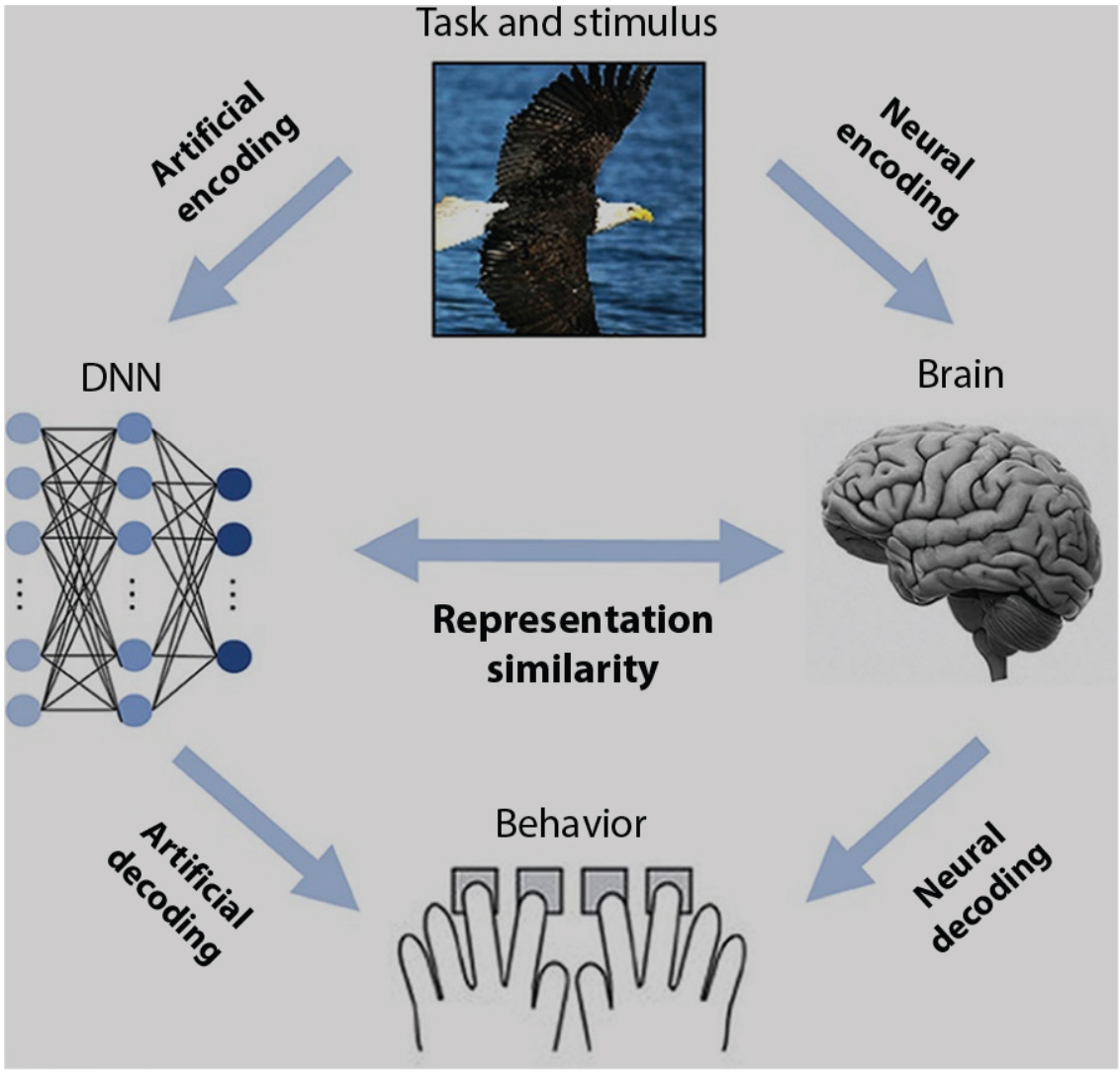



**Figure 19.4** Neural encoding and decoding processes, where sensory stimuli are transformed into neural activity patterns and vice versa [10].

# 19.3 How Neuroscience Can Help in Understanding Intelligent Automation Systems (IAS)

## 19.3.1 Neuroscientific Insights into Cognitive Processes

### Understanding Human Perception and Attention for AI Design

The human brain's remarkable abilities in perception and attention have long been a source of inspiration for the development of AI systems. Neuroscientific research has provided valuable insights into the neural mechanisms underlying these cognitive processes, which can inform the design of more effective and human-centric AI solutions. For example, studies on visual perception have revealed the intricate ways in which the brain processes and integrates sensory information, from the early stages of feature detection to the higher-level processing of objects and scenes. By emulating these neural mechanisms, AI systems can be trained to perceive and understand their environments more accurately, akin to how humans navigate their surroundings [11].

Similarly, research on attention has shed light on the brain's ability to selectively focus on relevant information while filtering out distractions, a crucial skill for efficient decision-making and task completion. Computational models of visual attention, such as saliency-based approaches and deep learning-based attention mechanisms, have been developed to mimic the brain's attentional processes and enhance the performance of AI systems in areas like object recognition, scene understanding, and robot navigation [12, 13]. By incorporating these insights into the design of AI-powered IAS, researchers can enhance their capacity to prioritize important information, allocate computational resources effectively, and respond to dynamic environments in a more human-like manner. Figure 19.5 shows the development of advanced AI learning algorithms inspired neuroscience.

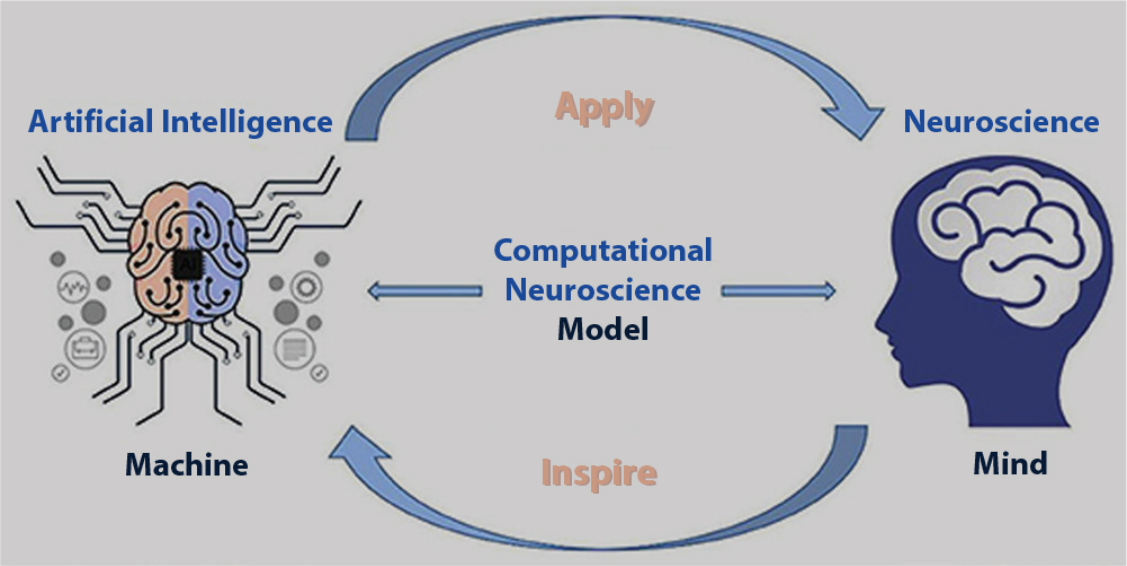



**Figure 19.5** Illustration of how neuroscientific insights into memory formation and learning can inspire the development of advanced AI learning algorithms [15].

### Memory Mechanisms and Learning Algorithms in AI Systems

The human brain's ability to store, recall, and update information is another area where neuroscience has provided valuable insights for the development of IAS. Researchers have studied the neural underpinnings of memory formation, consolidation, and retrieval, revealing the crucial roles of synaptic plasticity, neurogenesis, and neural network dynamics. These findings have inspired the creation of AI learning algorithms that mimic the brain's memory mechanisms, such as deep neural networks and reinforcement learning.

By incorporating these biologically inspired approaches, IAS can develop more robust and adaptive memory capabilities. For instance, deep neural networks can learn hierarchical representations of data, similar to the way the brain processes sensory information, while reinforcement learning algorithms can adapt their behavior through trial-and-error, akin to how humans and animals learn from experience. Additionally, research on memory consolidation and replay mechanisms has informed the development of AI systems that can continually learn and update their knowledge, without catastrophically forgetting previous information [14]. These advancements in biologically inspired learning algorithms have enabled IAS to become more flexible, adaptable, and capable of learning complex tasks over time, much like the human brain.

## 19.3.2 Brain-Inspired Algorithms and Models

### Mimicking Neural Networks for Efficient AI Solutions

The structure and function of biological neural networks have been a profound source of inspiration for the development of artificial neural networks (ANNs), a fundamental component of many AI systems. Bio-inspired neural network design is represented in Figure 19.6. Neuroscientific research has revealed the intricate details of how neurons and their interconnections process and transmit information, from the propagation of action potentials to the modulation of synaptic strengths. By mimicking these neural mechanisms, AI researchers have been able to create powerful ML algorithms that can tackle a wide range of problems, from image recognition to natural language processing [16].

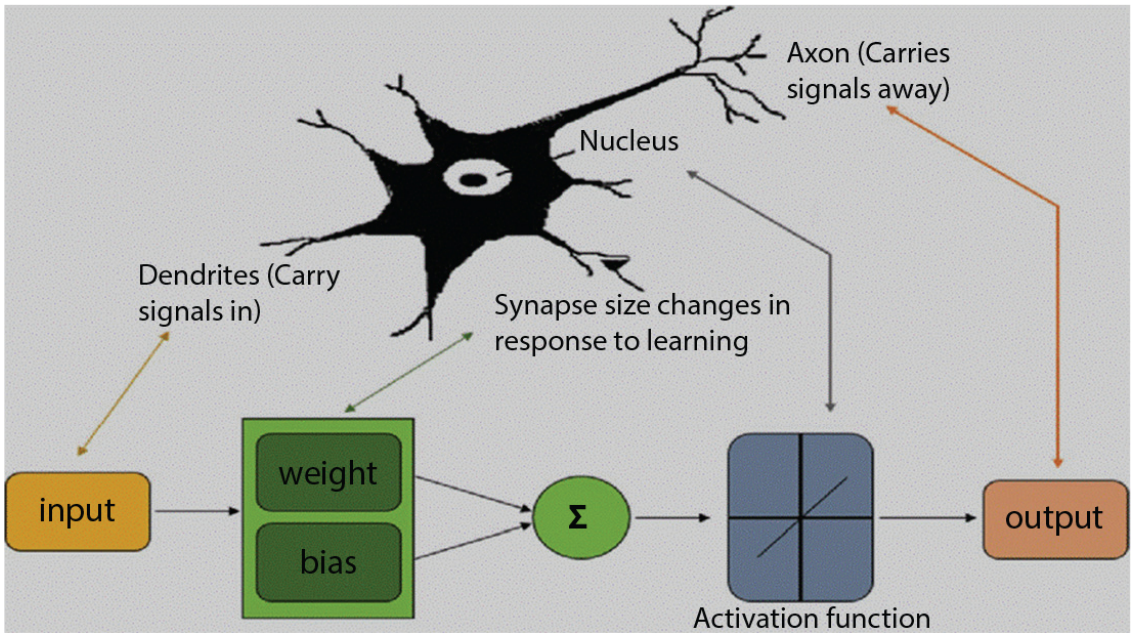



**Figure 19.6** Bio-inspired neural network design [18].

Moreover, the incorporation of biologically inspired principles, such as local connectivity, lateral inhibition, and hierarchical organization, has led to the development of more efficient and scalable AI architectures that can rival and even surpass human performance on specific tasks [17]. For example, the success of deep convolutional neural networks in image recognition tasks can be attributed to their ability to capture the hierarchical structure of visual information, much like the visual cortex in the human brain. Similarly, the attention mechanisms used in transformer-based models for natural language processing are inspired by the

brain's selective focus on relevant information. By continuing to draw inspiration from neuroscience, AI researchers can create even more powerful and efficient IAS that can seamlessly interact with and assist humans in a wide range of applications.

### Neuromorphic Computing Approaches for IAS

Building on the insights gained from neuroscience, the field of neuromorphic computing has emerged as a promising approach for the development of IAS. Neuromorphic computing aims to create hardware and software systems that mimic the brain's energy-efficient and parallel information-processing capabilities. By incorporating fundamental neural principles, such as spike-based communication, synaptic plasticity, and event-driven processing, neuromorphic architectures can offer significant advantages in terms of power consumption, real-time performance, and fault tolerance compared to traditional von Neumann-based computing systems.

The development of neuromorphic chips and neuromorphic software frameworks has paved the way for the integration of brain-inspired computing into IAS, enabling more efficient, adaptive, and autonomous decision-making and control. For example, neuromorphic sensors and processors can be used in robotic and autonomous systems to perceive and respond to their environments in a more energy-efficient and human-like manner, with potential applications in areas such as navigation, manipulation, and human-robot interaction [19]. Additionally, the inherent fault-tolerance and adaptability of neuromorphic architectures make them well-suited for deployment in harsh or unpredictable environments, where traditional computing systems may struggle to maintain reliable performance.

By further integrating neuroscientific insights into the design and development of IAS, researchers can create AI solutions that are more aligned with human cognitive processes, more adaptable to changing environments, and better equipped to collaborate with humans in a seamless and intuitive manner. The continued cross-pollination of neuroscience and AI holds great promise for advancing the field of intelligent automation and

unlocking new capabilities that can benefit a wide range of industries and applications.

## 19.4 Application Areas of Neuroscience in IAS

### 19.4.1 Healthcare and Medicine

#### Brain-Computer Interfaces for Rehabilitation and Assistive Technologies

The insights gained from neuroscience have been instrumental in the development of BCIs, which aim to directly connect the human brain to external devices and systems. In the healthcare and medical domain, BCIs have shown tremendous potential for rehabilitation and assistive technology. By decoding neural signals associated with movement intentions, BCIs can enable individuals with motor disabilities to control prosthetic limbs, wheelchairs, or computer interfaces, restoring their independence and improving their quality of life.

Furthermore, neuroscientific research on neural plasticity and motor learning has informed the design of BCI-powered rehabilitation systems that can facilitate the recovery of function after neurological injuries or diseases, such as stroke, spinal cord injury, or Parkinson's disease [20]. By integrating real-time neural feedback, these systems can guide patients through targeted rehabilitation exercises, promoting the reorganization of brain circuits and the relearning of lost motor skills. The synergy between neuroscience and intelligent automation in this domain has the potential to revolutionize the field of neurorehabilitation and assistive technologies. Figure 19.7 shows the bio-computer interface.

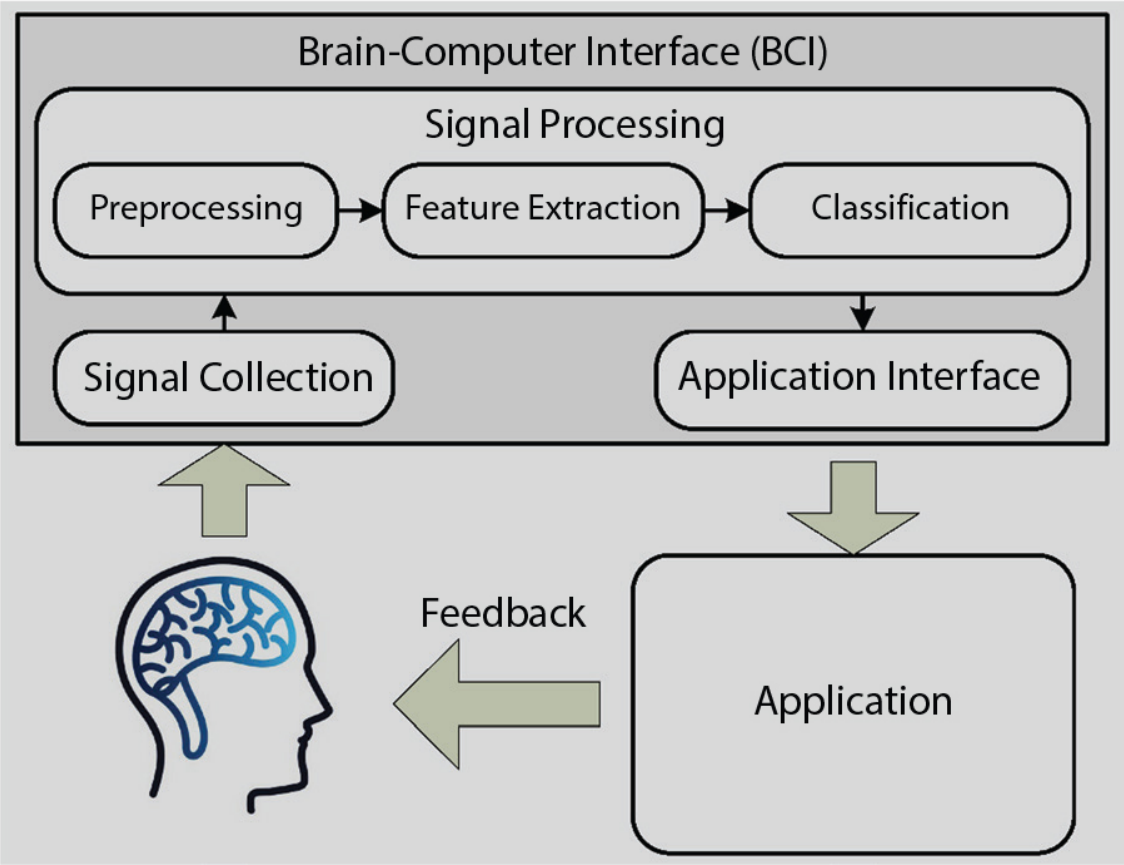



**Figure 19.7** Brain-computer interface system [21].

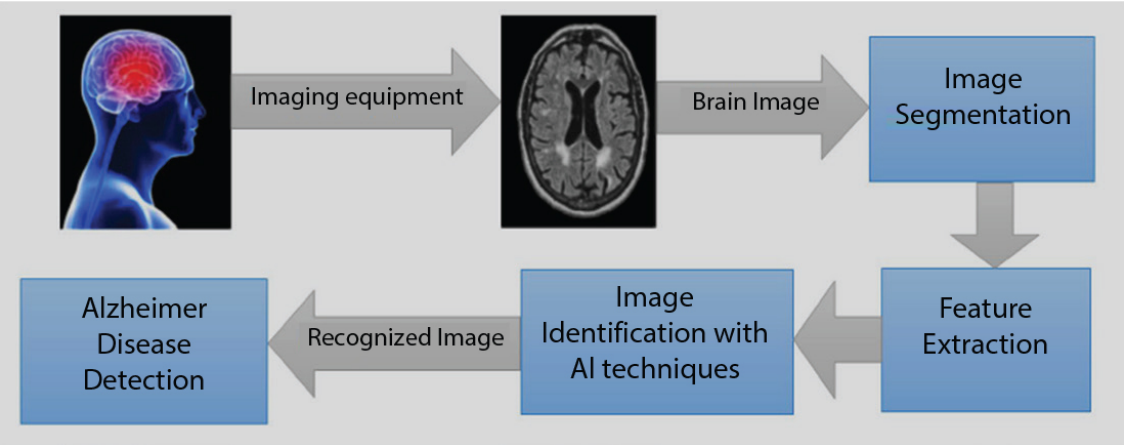



**Figure 19.8** Neuroscience-informed predictive analytics for disease diagnosis and treatment optimization in healthcare [23].

### Predictive Analytics for Disease Diagnosis and Treatment

Neuroscience has also provided valuable insights for the development of IAS in the field of healthcare and medicine, particularly in the area of disease diagnosis and treatment. By leveraging neuroscientific knowledge about the neural mechanisms underlying various neurological and psychiatric disorders, researchers have been able to develop advanced predictive analytics and decision support systems to aid clinicians [22].

For example, ML algorithms trained on neuroimaging data, such as functional magnetic resonance imaging (fMRI) or electroencephalography (EEG), can detect subtle patterns in brain activity that may be indicative of conditions like Alzheimer's disease, depression, or schizophrenia, often before the onset of overt symptoms. Additionally, neuroscience-informed computational models of disease progression can help predict the trajectory of neurological and psychiatric disorders, enabling more personalized and proactive treatment strategies. By integrating these IAS into clinical workflows, healthcare providers can potentially improve disease diagnosis, optimize treatment plans, and better allocate limited resources, ultimately leading to better patient outcomes. Figure 19.8 shows a neuroscience informed model for disease diagnosis.

### 19.4.2 Finance and Economics

#### Automated Trading Systems and Market Analysis

Neuroscience has also provided insights that have influenced the development of IAS in the finance and economics domains. One notable application is the use of neuroscientific knowledge in the design of automated trading systems and market analysis tools.

Researchers have leveraged findings from neuroeconomics, a field that combines neuroscience and economics, to better understand the neural underpinnings of decision-making processes in financial markets. By analyzing patterns of brain activity associated with risk-taking, reward processing, and emotional responses, scientists have been able to develop computational models that can simulate and predict human trading behavior more accurately. These insights have then been incorporated into IAS that can autonomously execute trading strategies, monitor market trends, and make investment decisions in a more informed and efficient manner.

Furthermore, neuroscience-inspired techniques, such as anomaly detection and changepoint analysis, have been employed in IAS to identify potential market anomalies, financial fraud, or other irregularities that may signal impending market fluctuations. By integrating these capabilities,

IAS can potentially enhance the performance and risk management of financial institutions, as well as provide valuable decision support for individual investors.

#### Neuroeconomic Models for Decision-Making

In addition to automated trading systems, neuroscience has also influenced the development of IAS in the field of economics and decision-making. Neuroeconomic models, which combine insights from neuroscience, psychology, and economics, have been used to create more realistic and predictive simulations of human economic behavior.

By studying the neural mechanisms underlying decision-making processes, such as value computation, risk assessment, and social interactions, researchers have been able to develop computational models that can more accurately capture the cognitive and emotional factors that drive economic decisions [24, 25]. These neuroeconomic models have then been integrated into IAS that can simulate complex economic scenarios, optimize decision-making strategies, and provide insights for policymakers and businesses.

For example, IAS based on neuroeconomic principles have been used to model consumer behavior, predict market trends, and design effective marketing strategies. Additionally, these systems can be employed to analyze the impact of economic policies, such as taxation or regulation, on individual and organizational decision-making, enabling more informed and evidence-based policymaking.

### 19.4.3 Manufacturing and Industry

#### Robotic Process Automation (RPA) in Production and Logistics

The application of neuroscience in IAS has also been influential in the manufacturing and industrial sectors. One area where this integration has proven particularly useful is in the development of robotic process automation (RPA) systems for production and logistics.

Neuroscientific research on motor control, sensory processing, and task learning has informed the design of more sophisticated and adaptable robotic systems that can seamlessly collaborate with human workers. By mimicking the neural mechanisms underlying human perception, decision-making, and motor skills, RPA systems can be trained to perform a wide range of tasks, from material handling and assembly to quality inspection and warehouse management, with increased efficiency, precision, and safety.

Furthermore, the integration of neuroscience-inspired learning algorithms, such as reinforcement learning and deep neural networks, has enabled RPA systems to adapt to changing environments, learn from experience, and continuously improve their performance over time, much like the human brain. This level of adaptability and autonomy is crucial in dynamic manufacturing and logistics settings, where RPA systems can optimize workflows, reduce errors, and enhance overall productivity.

### Human-Robot Collaboration in Manufacturing Environments

In addition to RPA, neuroscience has also played a significant role in the development of IAS designed for human-robot collaboration in manufacturing environments. By understanding the neural mechanisms underlying human perception, cognition, and social interaction, researchers have been able to create robotic systems that can more effectively and intuitively work alongside human workers [27, 28]. Figure 19.9 shows intelligent automaton system in industry. Figure 19.10 shows human-robot collaboration in manufacturing environments, highlighting the seamless interaction between humans and robots.

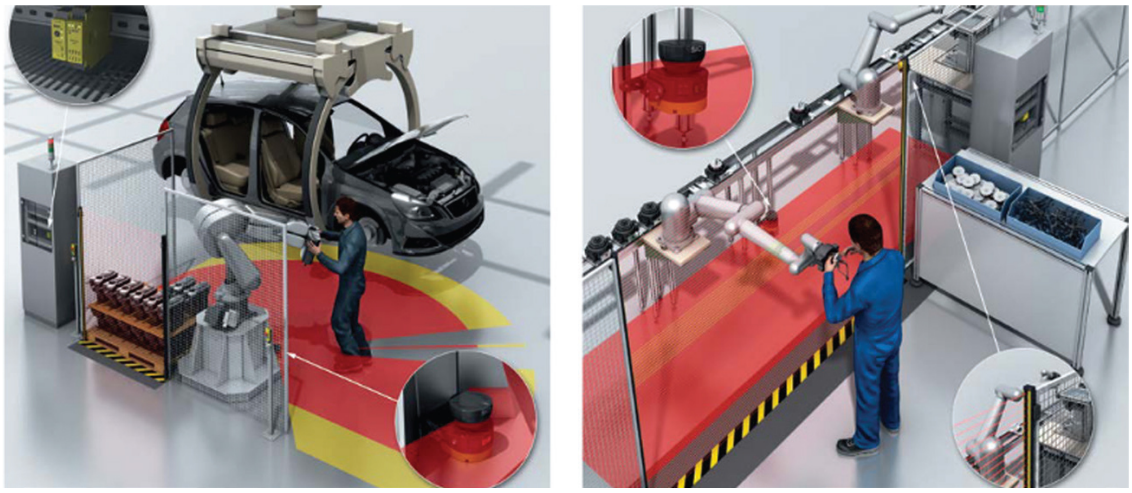


**Figure 19.9** Intelligent automation in industrial settings [26].

For example, neuroscience-inspired techniques for human intention recognition, based on the analysis of neural signals or observable behaviors, have enabled robotic systems to anticipate and respond to the needs and actions of their human counterparts in real time. This enhanced human-robot communication and coordination can lead to increased productivity, improved safety, and greater job satisfaction for human workers.

Furthermore, the incorporation of neuroscientific insights into the design of robotic interfaces and control systems has resulted in more natural and intuitive interactions, reducing the cognitive load and training required for human operators to work effectively with IAS. By fostering a seamless and collaborative relationship between humans and robots, these neuroscience-informed approaches can unlock new possibilities for the future of intelligent manufacturing and industrial automation.

The integration of neuroscience into the development of IAS has had a profound impact across various application domains, from healthcare and finance to manufacturing and industry. By drawing insights from the human brain's remarkable capabilities, researchers and engineers have been able to create AI-powered solutions that are more aligned with human cognition, more adaptable to dynamic environments, and better equipped to collaborate with humans in seamless and intuitive ways. As the cross-pollination of neuroscience and intelligent automation continues, we can expect to see even more innovative and transformative applications that can positively impact our lives and society.

## 19.5 Connecting IAS and Neuroscience

### 19.5.1 Neurofeedback and Brain-Computer Interfaces (BCIs)

Neurofeedback and BCIs represent a direct integration of neuroscience and IAS, with numerous applications across various domains. Neurofeedback involves the real-time monitoring and visualization of an individual's brain activity, allowing them to learn how to voluntarily control their neural processes. This technique has been widely explored in the healthcare and rehabilitation field, where it has shown promise in the treatment of conditions such as attention-deficit/hyperactivity disorder (ADHD), anxiety, depression, and even neurological disorders like Parkinson's disease.

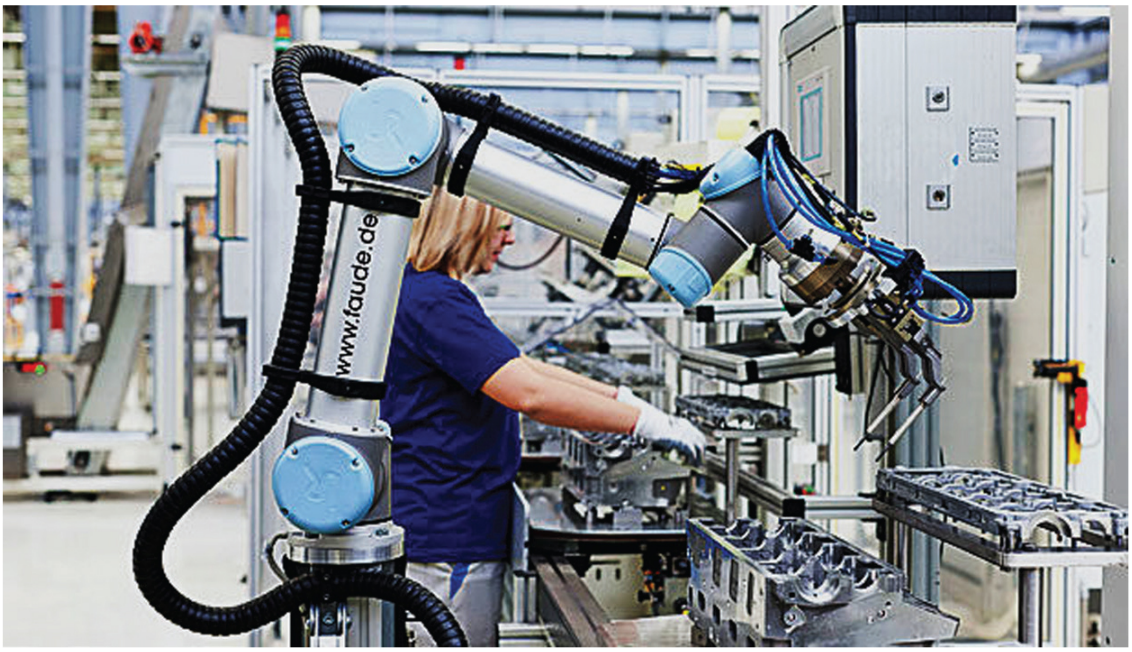



**Figure 19.10** Human-robot collaboration in manufacturing environments [31].

By providing individuals with immediate feedback on their brain activity, neurofeedback enables them to develop self-regulation strategies, leading to improvements in cognitive, emotional, and motor functions. IAS play a crucial role in neurofeedback, as they are responsible for the acquisition, processing, and presentation of neural signals, as well as the delivery of personalized feedback and training protocols.

Similarly, BCIs leverage the direct communication between the brain and external devices, enabling individuals to control or interact with technology using their neural activity. IAS are essential components of BCIs, as

they are responsible for decoding neural signals, interpreting user intentions, and translating them into commands for various applications, such as assistive technologies, gaming, and rehabilitation.

The integration of neuroscience and intelligent automation in the context of neurofeedback and BCIs has led to significant advancements in fields like healthcare, gaming, and neurorehabilitation, empowering individuals to harness the power of their own brain activity to improve their well-being, regain lost functionalities, and interact with the digital world in novel ways.

### 19.5.2 Cognitive Modeling and Human-Machine Interaction

The insights gained from neuroscience have also been instrumental in the development of more effective and intuitive human-machine interaction (HMI) systems. By understanding the cognitive processes underlying human perception, attention, decision-making, and learning, researchers have been able to design IAS that can better adapt to and collaborate with human users.

Cognitive modeling, which aims to computationally represent the mechanisms of human cognition, has been a key approach in this endeavor. By integrating neuroscientific knowledge into these models, researchers have created IAS that can more accurately predict and respond to human behavior, leading to improved interface design, task allocation, and overall system performance. Figure 19.11 shows the basic conceptual diagram of Brain -Computer Interface and Figure 19.12 represent the conceptual representation of human computer interaction.

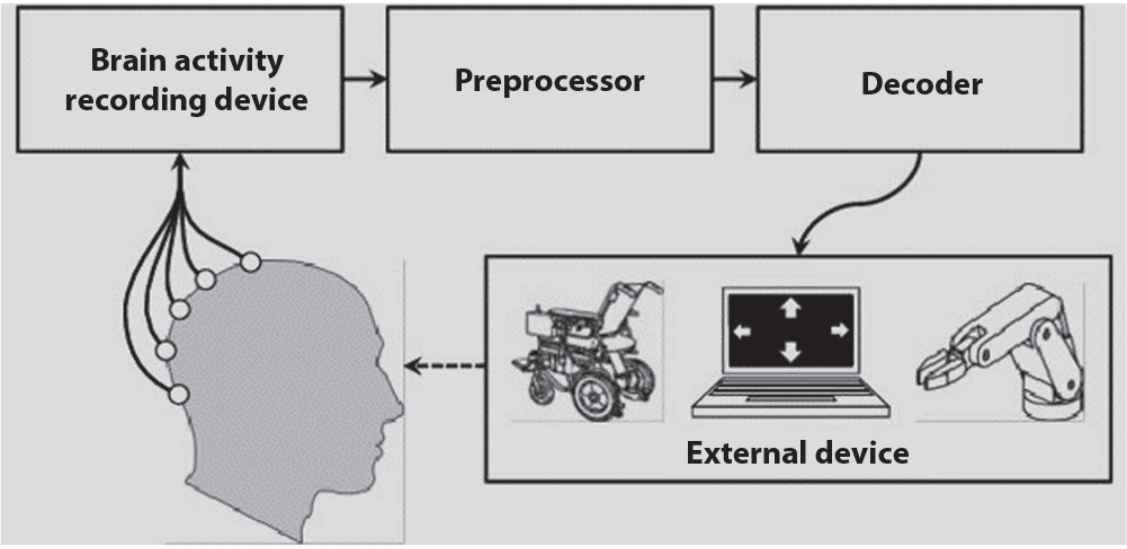



**Figure 19.11** Brain-computer interfaces (BCIs) [32].

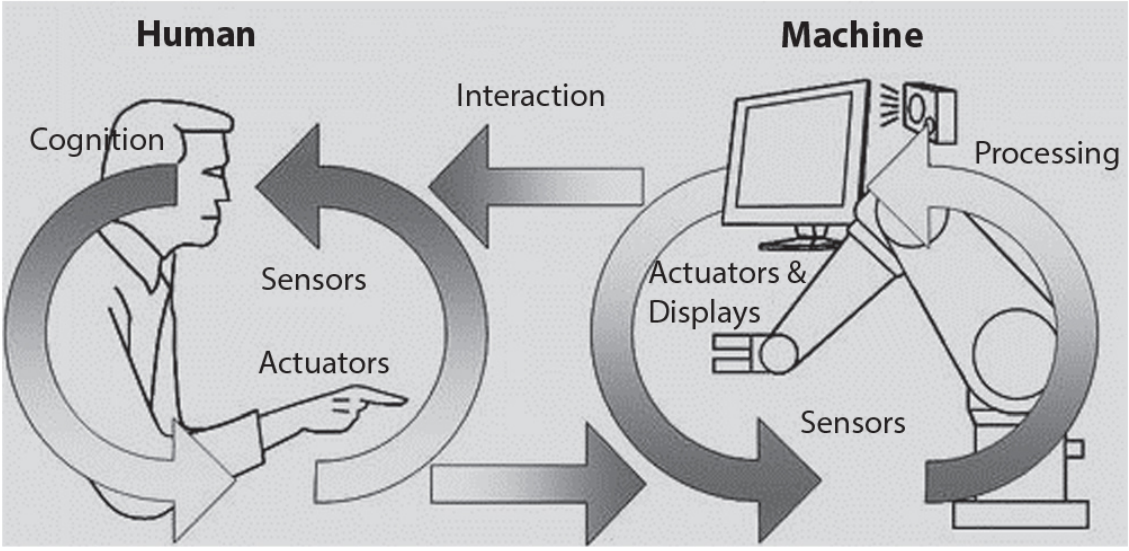



**Figure 19.12** Human-machine interaction [33].

Furthermore, the development of adaptive automation systems, which can dynamically adjust their level of autonomy and support based on the user's cognitive state and task demands, has been greatly influenced by neuroscience. By monitoring physiological and neural signals, such as brain activity, eye movements, and task performance, these systems can detect changes in the user's cognitive load, vigilance, and workload, and then adapt their level of assistance and automation accordingly. This integration of neuroscience and intelligent automation can lead to enhanced human-machine collaboration, improved task performance, and reduced cognitive fatigue in complex, dynamic environments.

### 19.5.3 Neuroergonomics

The field of neuroergonomics, which combines neuroscience and ergonomics, has also played a crucial role in the integration of IAS and neuroscience. Neuroergonomics focuses on understanding the neural mechanisms underlying human performance, cognition, and behavior in the context of human-machine systems, with the goal of enhancing safety, efficiency, and user experience.

By leveraging neuroscientific methods and insights, neuroergonomics researchers have been able to develop IAS that can better support and augment human capabilities. For example, the analysis of neural signals associated with attention, mental workload, and fatigue can inform the design of IAS that can detect and mitigate these cognitive states, reducing the risk of human errors and improving overall system performance [29].

Furthermore, neuroscientific findings on sensory processing, motor control, and ergonomic design principles have been instrumental in the development of IAS that can seamlessly integrate with human operators, leading to enhanced human-machine teaming and improved safety in domains such as transportation, manufacturing, and healthcare. By continuing to bridge the gap between neuroscience and intelligent automation, neuroergonomics can play a pivotal role in creating more human-centric and optimized interactions between humans and intelligent systems.

## 19.6 Challenges and Future Directions

### 19.6.1 Ethical and Social Implications

#### Privacy Concerns in Neuro-data Collection and Usage

As the integration of neuroscience and IAS becomes more prevalent, there are growing concerns about the ethical and social implications of these technologies. One significant challenge is the issue of privacy surrounding the collection and use of neuro data, which refers to the information gathered from the brain and nervous system.

The ability to access and analyze individuals' neural activity raises concerns about the potential for misuse, such as unauthorized monitoring, manipulation, or discrimination. Policymakers and researchers must address these privacy concerns by developing robust data governance frameworks and regulations that protect the privacy and autonomy of individuals while still enabling beneficial applications of neurotechnology.

Additionally, there are ethical considerations around the informed consent and transparency of neuro-data collection, as well as the potential for these technologies to be used for covert or non-consensual purposes. Addressing these challenges will be crucial in ensuring the responsible development and deployment of IAS that leverage neuroscientific insights.

**Bias and Fairness in Intelligent Automation Systems**

Another significant challenge in the integration of neuroscience and IAS is the potential for the introduction of bias and unfairness. Neuroscientific data and models can be influenced by various biases, such as demographic, cultural, or environmental factors, which may then be reflected in the design and outputs of IAS.

For example, if the neural data used to train ML algorithms is not representative of diverse populations, the resulting systems may exhibit biases in their perception, decision-making, or resource allocation, potentially leading to discriminatory outcomes. Addressing these biases will require a multifaceted approach, including the diversification of research participants, the development of debiasing techniques, and the implementation of rigorous testing and auditing procedures.

Additionally, the ethical allocation of benefits and burdens in IAS, particularly those that leverage neuroscientific insights, must be carefully considered to ensure fairness and equal access. Ongoing collaboration between ethicists, policymakers, and technical experts will be crucial in navigating these complex social and ethical challenges.

## 19.6.2 Technical Challenges

### Interpretable and Explainable AI

One of the significant technical challenges in the integration of neuroscience and IAS is the need for interpretable and explainable AI (XAI). Many of the AI algorithms and models inspired by neuroscience, such as deep neural networks, can be highly complex and opaque, making it difficult to understand the reasoning behind their decisions and outputs.

This lack of transparency and interpretability can be problematic, particularly in highstakes applications like healthcare, finance, or public policy, where stakeholders and end-users require a clear understanding of how the system arrived at its conclusions. Developing XAI techniques that can provide meaningful explanations of the inner workings of neuroscience-inspired AI systems will be crucial in building trust and ensuring the responsible deployment of these technologies.

Additionally, the integration of neuroscientific insights into the design of XAI systems may lead to new approaches that better align with human cognitive processes and decision-making. By drawing inspiration from the brain's ability to reason, learn, and explain, researchers can work toward creating AI systems that are more transparent, accountable, and trustworthy.

### Closing the Gap between Neural Networks and the Human Brain

Another significant technical challenge in the integration of neuroscience and IAS is the difficulty in fully replicating the complexity and capabilities of the human brain using current AI and computational approaches.

While artificial neural networks have been inspired by the structure and function of biological neural networks, they still fall short in capturing the full breadth and depth of neural processing, learning, and adaptation observed in the human brain [30]. Factors such as the brain's hierarchical organization, dynamic synaptic plasticity, neuromodulation, and the interplay between different brain regions remain challenging to model and implement in AI systems.

Closing this gap will require continued advancements in neuroscience research, as well as the development of more biologically plausible and scalable computational frameworks that can better emulate the brain's information-processing and learning mechanisms. Breakthroughs in areas like neuromorphic computing, spiking neural networks, and reinforcement learning may play a crucial role in bridging this divide and unlocking new levels of intelligence and adaptability in AI systems.

### 19.6.3 Future Trends

#### Neuromorphic Computing and Brain-Inspired AI

One of the promising future directions in the integration of neuroscience and IAS is the continued development of neuromorphic computing and brain-inspired AI.

Neuromorphic computing aims to create hardware and software architectures that mimic the brain's energy-efficient and parallel information-processing capabilities, offering significant advantages in terms of power consumption, real-time performance, and fault tolerance compared to traditional von Neumann-based computing systems.

By directly incorporating neuroscientific principles, such as spike-based communication, synaptic plasticity, and event-driven processing, neuromorphic systems can enable more adaptive, robust, and human-like intelligent automation solutions. As these technologies mature, they have the potential to revolutionize a wide range of applications, from robotics and autonomous systems to edge computing and distributed sensor networks.

In parallel, the development of brain-inspired AI architectures, which leverage neuroscientific insights to create more powerful and efficient ML algorithms, will continue to be a focus of research and innovation. By drawing inspiration from the brain's remarkable capabilities in areas like perception, reasoning, and adaptation, researchers can work toward creating AI systems that can surpass the current limitations of conventional approaches and unlock new frontiers of intelligent automation.

#### Integrating Neuroscientific Insights into AI Development

Another key future direction in the integration of neuroscience and IAS is the continued efforts to incorporate neuroscientific insights more seamlessly into the development and deployment of AI-powered technologies.

As our understanding of the brain's structure, function, and information-processing mechanisms continues to evolve, researchers and developers will seek to leverage these insights to create more robust, adaptable, and human-centric IAS. This may involve the integration of neuroscientific principles into the design of ML algorithms, the development of more biologically plausible neural network architectures, and the incorporation of neurocognitive models into the decision-making and control frameworks of intelligent systems.

Furthermore, the real-time monitoring and analysis of neural activity, coupled with the advancements in neurotechnology and BCIs, may enable IAS to adapt and respond to the cognitive and emotional states of their human users, leading to more seamless and intuitive human-machine collaboration.

By continuously bridging the gap between neuroscience and AI, researchers and developers can work toward creating IAS that are more aligned with human cognition, more responsive to human needs, and better equipped to address the complex challenges facing our society.

As the integration of neuroscience and IAS continues to evolve, it will be crucial to address the ethical, social, and technical challenges that arise. By proactively considering the privacy implications, bias and fairness concerns, and the need for interpretable and explainable AI, researchers and developers can work toward creating IAS that are not only technologically advanced but also socially responsible and aligned with human values. Additionally, the ongoing efforts to bridge the gap between neural networks and the human brain, as well as the advancements in neuromorphic computing and brain-inspired AI, hold the promise of unlocking new frontiers of intelligent automation that can positively impact various domains, from healthcare and finance to manufacturing and beyond.

## 19.7 Conclusion

In conclusion, this chapter has provided a comprehensive overview of the intersection between neuroscience and IAS. We began by defining neurosciences and IAS, highlighting the significance of bridging these disciplines. Fundamental concepts of neuroscience were explored, including neurons, neural networks, brain structure, and cognitive processes such as perception, attention, memory, and decision-making. We delved into how insights from neuroscience can enhance our understanding of IAS, from designing AI systems that mimic human cognitive processes to leveraging brain-inspired algorithms and models. Moreover, we discussed various application areas where neuroscience can inform the development of IAS, spanning healthcare, finance, economics, manufacturing, and industry. Examples include BCIs in healthcare, automated trading systems in finance, and robotic process automation in manufacturing.

The chapter also emphasized the importance of connecting IAS and neuroscience through neurofeedback, BCIs, cognitive modeling, and neuroergonomics. These approaches can lead to the development of more intuitive interfaces, adaptive automation systems, and enhanced human performance and safety. However, several challenges and future directions were identified, including ethical and social implications, technical challenges in creating interpretable AI, and the need to close the gap between neural networks and the human brain. Nonetheless, the vision for the future of neurosciences and IAS is promising, with opportunities for continued research and collaboration between neuroscientists and AI developers.

Future research should concentrate on designing more sophisticated brain-inspired AI architectures, improving real-time adaptive systems with improved neurofeedback, and identifying potential applications of neuroergonomics across industries. Addressing AI interpretability and closing the gap between artificial and biological neural networks are still key concerns. As these domains converge, we expect revolutionary applications that will redefine human-technology interactions and have a profound impact on industries such as healthcare, education, and others.

## References


1. Floreano, D., Husbands, P., Nolfi, S., Evolutionary Robotics. In: Siciliano, B., Khatib, O. (eds.) Springer Handbook of Robotics, Springer, Berlin, Heidelberg, 2008. https://doi.org/10.1007/978-3-540-30301-5_62.
2. Groover, M.P., *Automation, production systems, and computer-integrated manufacturing*, Pearson Education India, USA, 2016.
3. http://www.robots.ox.ac.uk
4. Dayan, P. and Abbott, L.F., Theoretical neuroscience: computational and mathematical modeling of neural systems. *J. Cognit. Neurosci.*, 15, 1, 154–155, 2003.
5. Gerstner, W., *et al.*, *Neuronal dynamics: From single neurons to networks and models of cognition*, Cambridge University Press, United Kingdom, 2014.
6. https://vajiramandravi.com/quest-upsc-notes/artificial-neural-network
7. Kandel, E.R., *et al.* (Eds.), *Principles of neural science*, vol. 4, McGraw-hill, New York, 2000.
8. Holtmaat, A. and Svoboda, K., Experience-dependent structural synaptic plasticity in the mammalian brain. *Nat. Rev. Neurosci.*, 10, 9, 647–658, 2009.
9. Schwartz, A.B., *et al.*, Brain-controlled interfaces: movement restoration with neural prosthetics. *Neuron*, 52, 1, 205–220, 2006.
10. Chen, X., *et al.*, DNNBrain: a unifying toolbox for mapping deep neural networks and brains. *Front. Comput. Neurosci.*, 14, 580632, 2020.
11. Riesenhuber, M. and Poggio, T., Hierarchical models of object recognition in cortex. *Nat. Neurosci.*, 2, 11, 1019–1025, 1999.
12. Borji, A. and Itti, L., State-of-the-art in visual attention modeling. *IEEE Trans. Pattern Anal. Mach. Intell.*, 35, 1, 185–207, 2012.
13. Vaswani, A., *et al.*, Attention is all you need. *Adv. Neural Inf. Process. Syst.*, 30, 6000–6010, 2017.
14. Kumaran, D., Hassabis, D., McClelland, J.L., What learning systems do intelligent agents need? Complementary learning systems theory updated. *Trends Cognit. Sci.*, 20, 7, 512–534, 2016.
15. Li, J., *et al.*, Closed-loop iterations between neuroscience and artificial intelligence. *Front. Syst. Neurosci.*, 16, 1002095, 2022.
16. LeCun, Y., Bengio, Y., Hinton, G., Deep learning. *Nature*, 521, 7553, 436–444, 2015.

17. Krizhevsky, A., Sutskever, I., Hinton, G.E., Imagenet classification with deep convolutional neural networks. *Adv. Neural Inf. Process. Syst.*, 25, 84–90, 2012.
18. Sharifi, Y., *et al.*, Study of neural network models for the ultimate capacities of cellular steel beams. *Iran. J. Sci. Technol. Trans. Civ. Eng.*, 44, 2, 579–589, 2020.
19. Qiao, N., *et al.*, A reconfigurable on-line learning spiking neuromorphic processor comprising 256 neurons and 128K synapses. *Front. Neurosci.*, 9, 141, 2015.
20. Ang, K.K. and Guan, C., Brain–computer interface for neurorehabilitation of upper limb after stroke. *Proc. IEEE*, 103, 6, 944–953, 2015.
21. Peksa, J. and Mamchur, D., State-of-the-art on brain-computer interface technology. *Sensors*, 23, 13, 6001, 2023.
22. Widge, A.S., *et al.*, Treating refractory mental illness with closed-loop brain stimulation: progress towards a patient-specific transdiagnostic approach. *Exp. Neurol.*, 287, 461–472, 2017.
23. Subasi, A., Use of artificial intelligence in Alzheimer's disease detection, in: *Artificial intelligence in precision health*, pp. 257–278, 2020.
24. Kuhnen, C.M. and Knutson, B., The influence of affect on beliefs, preferences, and financial decisions. *J. Financ. Quant. Anal.*, 46, 3, 605–626, 2011.
25. Rangel, A., Camerer, C., Read Montague, P., A framework for studying the neurobiology of value-based decision making. *Nat. Rev. Neurosci.*, 9, 7, 545–556, 2008.
26. https://cdn.sick.com/media/docs/6/96/996/whitepaper_safe_robotics_en_im0072996.pdf
27. Villani, V., *et al.*, Survey on human–robot collaboration in industrial settings: Safety, intuitive interfaces and applications. *Mechatronics*, 55, 248–266, 2018.
28. Ajoudani, A., *et al.*, Progress and prospects of the human–robot collaboration. *Auton. Rob.*, 42, 957–975, 2018.
29. Aricò, P., *et al.*, Passive BCI beyond the lab: current trends and future directions. *Physiol. Meas.*, 39, 8, 08TR02, 2018.
30. Marblestone, A.H., Wayne, G., Kording, K.P., Toward an integration of deep learning and neuroscience. *Front. Comput. Neurosci.*, 10, 215943, 2016.

31. https://www.assemblymag.com/articles/91862-human-robot-collaboration-comes-of-age
32. Mora-Cortes, A., *et al.*, Language model applications to spelling with brain-computer interfaces. *Sensors*, 14, 4, 5967–5993, 2014.
33. Razzaq, M.A., *et al.*, A survey on user interfaces for interaction with human and machines. *Int. J. Adv. Comput. Sci. Appl.*, 8, 7, 2017.

## Note

1. *Corresponding author: bikram.on9cls@gmail.com